Climate Policy Uncertainty and Financial Markets

Evidence from Stock Prices, Return Volatility, and Trading Activity

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Motivation & Research Question

- Climate policy is subject to frequent **reversals**: Paris Accords withdrawal (2017), French fuel tax repeal, Australia coal phase-out abandonment.
- **Climate Policy Uncertainty (CPU)** affects firms' investment decisions (Dixit & Pindyck, 1994; Gavriilidis et al., 2026) and stock valuations (Present Value Theory).
- **Limited evidence** on:
  - VWAP-based returns
  - Realized volatility
  - Trading volume dynamics

Research Question

How does climate policy uncertainty affect stock returns, realized volatility, and trading volume for large U.S. firms?

Text Data for the CPU Index

Campaign Materials (APP)

- Source: American Presidency Project
- Period: 2016Q1–2025Q4
- Types: Executive Orders, Statements, Campaign Documents, Debates, etc.
- Filtered to climate-related documents using Gavriilidis (2023) dictionary

Legislative Texts (BILLS)

- Source: GovInfo Bulk Data (congress.gov)
- Congresses 114–119 (2015–2025)
- House & Senate bills + Joint Resolutions
- Same climate dictionary filter applied

Climate Dictionary (Gavriilidis et al., 2023)

Three tiers: trigrams (e.g. "cap and trade"), bigrams (e.g. "climate change", "carbon tax"), and unigrams (e.g. "epa", "co2", "methane").

Financial Data

- **Sample**: 30 companies of the Dow Jones Industrial Average (Jan 2017 – Dec 2025).
- **Stock prices**: Daily Volume Weighted Average Price (VWAP) from LSEG → monthly arithmetic returns.
- **Trading volume**: Daily data from Yahoo Finance → abnormal volume & first-difference log volume.
- **Control variable**: VIX index (CBOE) from FRED.

Financial Data

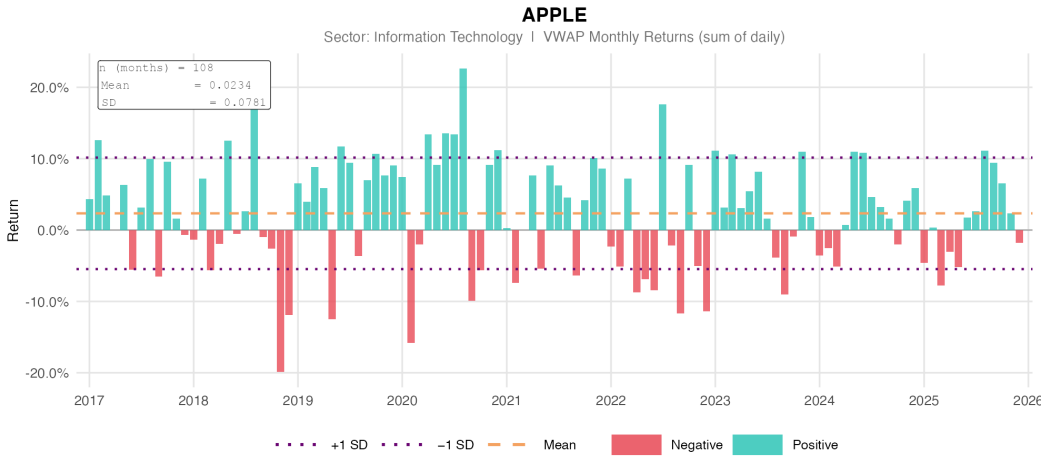
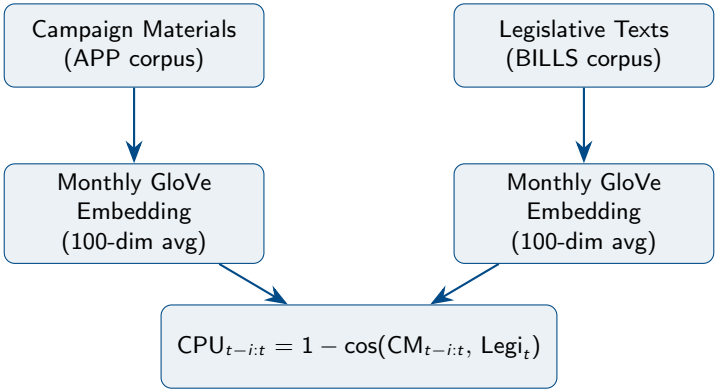


Figure: APPLE monthly VWAP returns over time

Building the CPU Index



- **High CPU** = campaign rhetoric *diverges* from legislative language.
- **Preferred specification**: CPU(t:t−2) — 3-month window for campaign materials.
- Robustness with $i \in \{0, 1, 2, 3\}$ shows results are not sensitive to the lag choice.

Spike Detection Methods

Method 1: Rolling Threshold

$$\text{Spike}_t = 1[\text{CPU}_t > \bar{\mu}_{12} + 1.5\,\hat{\sigma}_{12}]$$

Method 2: Local Maxima (Preferred)

$$\text{Spike}_t = 1[\text{CPU}_t > \text{CPU}_{t\pm1} \ \wedge \ \text{CPU}_t > \bar{\mu} + 1.5\sigma]$$

- Identifies *genuine peaks*, not elevated plateaus
- Less sensitive to window length

- Minimum **5-month gap** enforced between spikes (non-overlapping ± 2 event windows).

Stacked Event-Study Design

Following Baker et al. (2022) and Wing et al. (2024):

$$y_{i,e,h} = \alpha_i + \sum_{h \neq -2} \beta_h \, 1[\text{rel_time}_e = h] + \gamma \, X_{i,e,h} + \varepsilon_{i,e,h}$$

- $h \in \{-2, -1, 0, +1, +2\}$: symmetric event window around each spike.
- $h = -2$ is the **omitted baseline**.
- α_i: firm fixed effects; SE clustered at the firm level.
- Avoids TWFE bias from heterogeneous treatment effects.

The VIX Endogeneity Problem

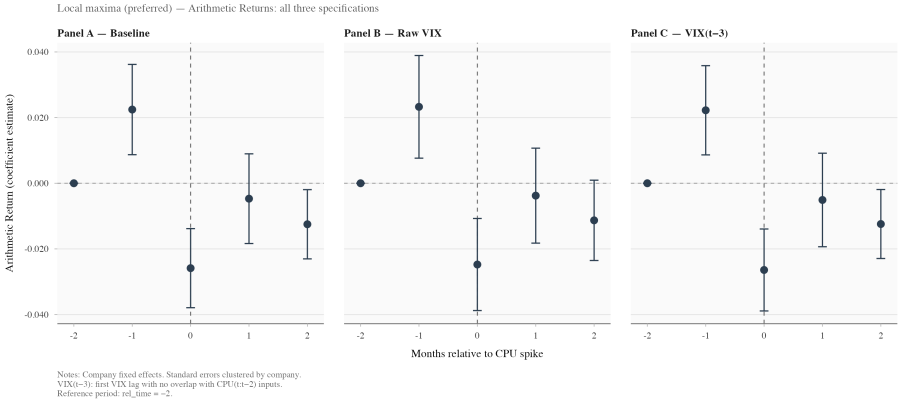
Key concern: Is VIX a valid control or a "bad control"?

- **Granger causality**: Neither CPU $\rightarrow$ VIX nor VIX $\rightarrow$ CPU is significant ($p > 0.14$).
- **VAR instantaneous causality**: $\chi^2(1) = 2.757$, $p = 0.097$
  $\Rightarrow$ CPU and VIX are *contemporaneously correlated* within the same month.
- **Cross-correlation**: Strongest at lag 0 (≈ -0.30); negligible at other lags.

Solution: VIX(t$-$3)

CPU(t:t$-$2) uses APP from months t, $t-1$, $t-2$ and BILLS at t.
$\Rightarrow$ VIX(t$-$3) is the **first lag with no overlap** with any CPU input month.
$\Rightarrow$ Fully predetermined by construction.

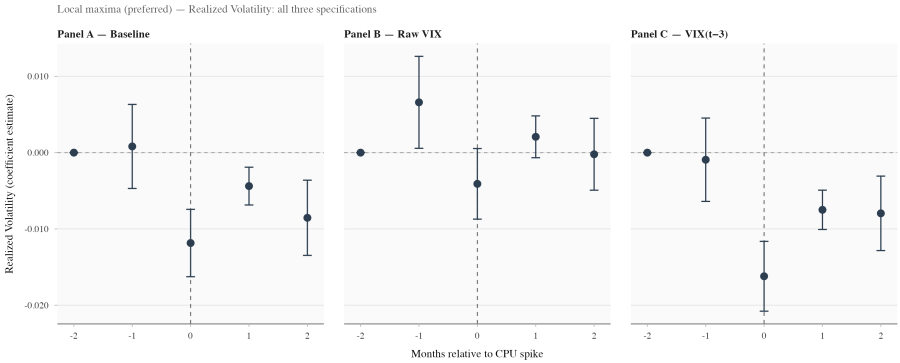
Result 1: Arithmetic Returns



Note

Stock returns are significantly positive in the month before the spike ($t = -1$), consistent with anticipation effects, and significantly negative in the spike month ($t = 0$), reflecting a higher risk of stock price crashes.

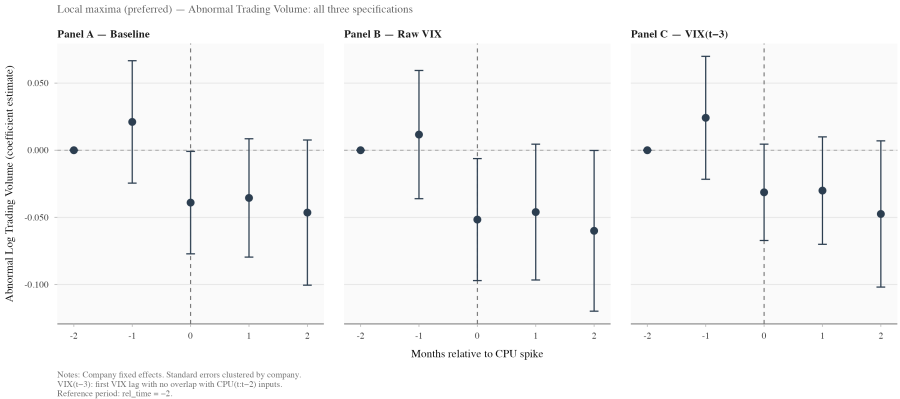
Result 2: Realized Volatility



Note

CPU spikes are followed by a significant and persistent decline in realized volatility, consistent with the wait-and-see mechanism in Bloom (2009).

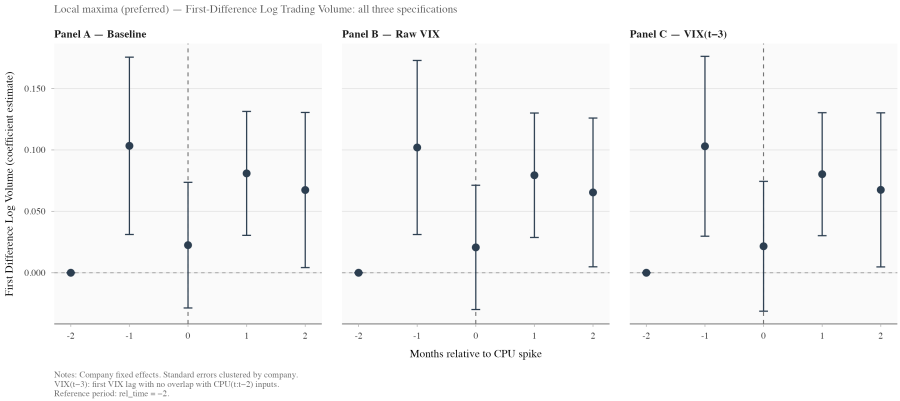
Result 3: Abnormal Trading Volume



Note

Trading volume reacts negatively to CPU shocks compared to the yearly mean, low significance so look at the First Difference for more information.

Result 4: First-Difference Log Volume



Note

Positive reaction of the First difference trading volume before and after the shock coherent with an anticipation effect combined with portfolio hedging.

Limitations & Future Research

Limitations:

- CPU index may partly capture **corpus-size effects**: months with high legislative output inflate measured divergence.
- Pre-trend coefficients β_{-1} are significant for returns and first-difference log volume $\Rightarrow$ the parallel trends assumption may not fully hold (or genuine anticipation).
- Sample limited to **30 Dow Jones firms**.

Thank you!

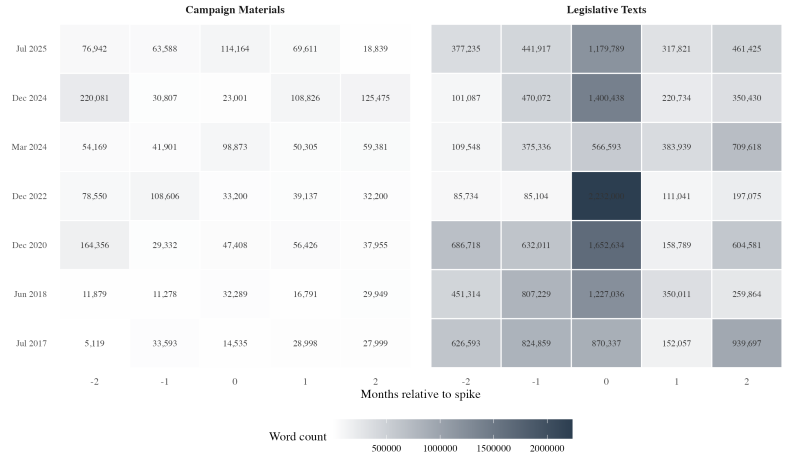
Questions & Discussion

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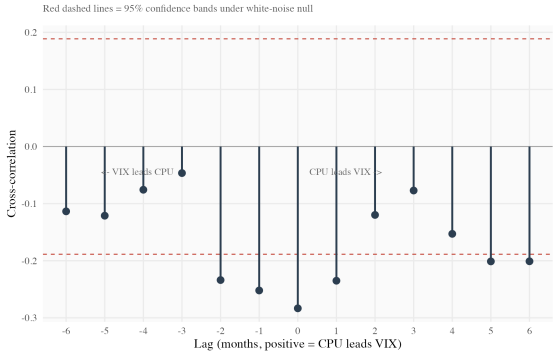
Appendix

Figure: Text volume (word count) around CPU spike episodes



Appendix

Figure: Cross-Correlation Function: CPU(t:$t-2$) and VIX.



Notes: Positive lags denote CPU leading VIX; negative lags denote VIX leading CPU. Red dashed lines indicate 95% confidence bands under the white-noise null ($\pm 1.96/\sqrt{n}$). Bars extending beyond the bands are statistically significant.